

PYTHON MINI PROJECT REPORT

Project Title

Belt Drive Transmission (Flat / V-Belt) Design Checker

Course Name: Semester-3: Python Programming

Class Name: Mechanical-3A

Branch Name: Mechanical Engineering / Automation & Robotics

Prepared By

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Submitted to

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CERTIFICATE

This certificate is proudly presented to the following students for successfully completing the Python Mini-Project entitled “Belt Drive Transmission (Flat / V-Belt) Design Checker” as part of their Diploma in Mechanical Engineering curriculum. This project work has been carried out under the guidance and supervision of the project guide.

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TABLE OF CONTENTS

CERTIFICATE	2
TABLE OF CONTENTS	3
1. INTRODUCTION	4
2. OBJECTIVE	4
3. METHODOLOGY	4
4. MECHANICAL ENGINEERING APPLICATION	4
5. PYTHON CODE	4
6. SCREENSHOTS	5
7. RESULTS AND DISCUSSIONS	5
8. CONCLUSION	5
9. REFERENCES	5
10. COURSE CERTIFICATES	6

1. INTRODUCTION

Belt drives are mechanical power-transmission systems used to transfer rotary motion and power between shafts. The project develops a simple web-based design checker for flat belts and V-belts. The user enters basic pulley, speed, centre-distance, friction and belt data, and the program calculates important design values such as belt speed, belt length, angle of wrap, tension ratio and power transmission. The aim is to reduce repetitive manual calculations and provide a quick engineering check.

2. OBJECTIVE

The main objectives are to: (1) provide a simple website interface for belt-drive calculations; (2) apply standard belt-drive formulae using Python; (3) calculate transmission quantities for flat and V-belt arrangements; (4) compare calculated power with a required/design power where applicable; and (5) clearly display results and basic design warnings.

3. METHODOLOGY

The project follows an input → validation → calculation → result-check workflow. The web page accepts pulley diameters, rotational speeds, centre distance, coefficient of friction, belt mass and other required values. Python performs the calculations using functions for belt speed, velocity ratio, belt length, wrap angle, tension ratio and power. The calculated values are returned to the web page in a readable result table. Flask can be used as the Python web framework, while HTML/CSS provides the front-end form.

Key formulae used:

- Belt speed: $v = \pi DN / 60$
- Velocity ratio (ideal, no slip): $N_1/N_2 = D_2/D_1$
- Open-belt length: $L = \pi(D+d)/2 + 2C + (D-d)^2/(4C)$
- Angle of wrap on small pulley: $\theta = \pi - 2 \sin^{-1}[(D-d)/(2C)]$
- Power transmitted: $P = (T_1 - T_2)v$
- Centrifugal tension: $T_c = mv^2$
- Flat-belt tension ratio: $T_1/T_2 = e^{\mu\theta}$

For V-belts, the tension-ratio relation includes the groove-angle effect; the exact expression depends on the belt/groove convention used.

4. MECHANICAL ENGINEERING APPLICATION

The project applies directly to machine design and power transmission. Belt drives are commonly used in fans, conveyors, pumps, machine tools and other rotating equipment. The checker helps a student or designer estimate whether a selected pulley arrangement can transmit the required power and whether basic geometric conditions are reasonable. It is intended as an educational design-checking tool and not as a replacement for manufacturer belt selection or a complete machine-design standard.

5. PYTHON CODE

```
from math import pi, asin, exp

def belt_speed(D_mm, rpm):
    return pi * (D_mm / 1000) * rpm / 60

def open_belt_length(D_mm, d_mm, C_mm):
    D, d, C = D_mm, d_mm, C_mm
    return pi*(D+d)/2 + 2*C + (D-d)**2/(4*C)

def wrap_angle(D_mm, d_mm, C_mm):
    return pi - 2*asin((D_mm-d_mm)/(2*C_mm))

def flat_belt_ratio(mu, theta):
    return exp(mu * theta)

def transmitted_power(T1, T2, v):
    return (T1 - T2) * v # watts

# Example:
D, d, C, N = 400, 200, 1000, 1440
v = belt_speed(D, N)
L = open_belt_length(D, d, C)
```

```
theta = wrap_angle(D, d, C)
ratio = flat_belt_ratio(0.30, theta)
print(v, L, theta, ratio)
```

6. SCREENSHOTS

The final website should include screenshots of the input form, calculated output and any validation/warning messages. Since the report template requires screenshots, the following sample representation can be used until actual website screenshots are inserted.

INPUT / OUTPUT FIELD	SAMPLE VALUE
Belt type	Flat belt
Driver pulley diameter D	400 mm
Driven pulley diameter d	200 mm
Centre distance C	1000 mm
Driver speed N	1440 rpm
Coefficient of friction μ	0.30
Belt speed v	30.16 m/s (approx.)
Open-belt length L	2631.86 mm (approx.)
Small-pulley wrap angle θ	2.94 rad (approx.)
Flat-belt tension ratio	2.42 (approx.)

Suggested screenshot captions: Figure 1 – Website input page; Figure 2 – Calculated belt-drive results; Figure 3 – Design warning/validation message.

7. RESULTS AND DISCUSSIONS

For the sample input, the calculator determines the belt speed, open-belt length and small-pulley wrap angle from the selected pulley geometry and speed. The tension ratio for the flat-belt example is obtained from the exponential friction relation. These outputs demonstrate how a web calculator can make repeated belt-drive calculations faster and easier to check. The result page can also flag invalid geometry, such as an impossible wrap-angle calculation or non-positive centre distance. Final belt selection still requires checking allowable belt tension, service factor, pulley limits, slip/creep and manufacturer data.

8. CONCLUSION

The Belt Drive Transmission (Flat / V-Belt) Design Checker provides a compact application of Python programming to a mechanical-engineering design problem. The project converts standard belt-drive formulae into a structured calculation workflow and presents the results through a website interface. The objectives of automating basic calculations, improving readability and providing preliminary design checks are achieved. The main limitation is that the tool is a preliminary educational checker; detailed belt selection should be verified against relevant standards and manufacturer catalogues.

9. REFERENCES

1. PSG College of Technology, Design Data Book, relevant sections on belt drives and power transmission.
2. R. S. Khurmi and J. K. Gupta, A Textbook of Machine Design, belt and rope drives chapter.
3. V. B. Bhandari, Design of Machine Elements, power transmission and belt-drive design.
4. Python Software Foundation, Python documentation.
5. Flask documentation for Python web applications.

10. COURSE CERTIFICATES

Attach relevant Python Programming / course completion certificates here.

Course Certificate 1

Course Certificate 2